Lines, Triangles & Polygons

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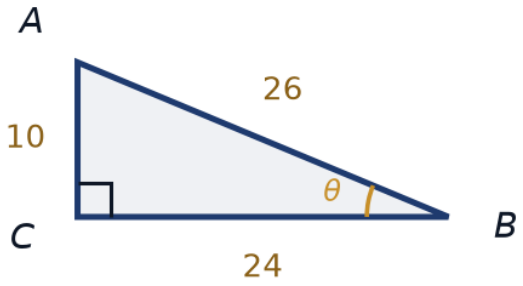
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Parallel lines and transversals

Two parallel lines are cut by a transversal, forming an angle of 105° on one line. Find the corresponding angle, the alternate interior angle, and the co-interior (same-side interior) angle on the other line.

2 [Foundational] • Target: Parallel lines and transversals

Two angles formed by a transversal crossing parallel lines are alternate exterior angles, so they are equal. If one measures $3x + 10$ and the other measures $5x - 30$, find x and the measure of the angle.



3 [Foundational] • Target: Parallel lines and transversals

Two parallel lines are cut by a transversal. One interior angle is $4x + 15$, and its corresponding angle on the other line is $6x - 25$. Find x and the angle measure.

4 [Foundational] • Target: Triangle angle sum and the exterior angle theorem

The angles of a triangle are in the ratio $3 : 5 : 7$. Find each angle.

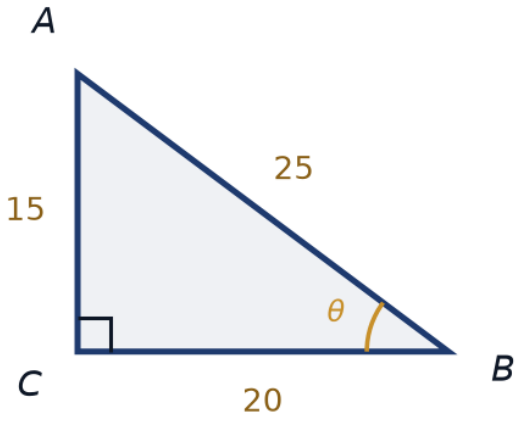
5 [Foundational] • Target: Triangle angle sum and the exterior angle theorem

In triangle ABC, $\angle A = 52^\circ$ and the exterior angle at C is 130°. Find $\angle B$.

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6 [Foundational] • Target: Triangle angle sum and the exterior angle theorem

An exterior angle of a triangle measures 115°, and one remote interior angle is 50°. Find the other remote interior angle and the third (adjacent) interior angle.



7 [Foundational] • Target: Triangle congruence criteria

State which congruence criterion (SSS, SAS, ASA, AAS, or RHS) proves triangles PQR and XYZ congruent, given $PQ = XY$, $\angle Q = \angle Y$, and $QR = YZ$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Triangle congruence criteria

Triangle DEF has $DE = 9$, $EF = 12$, $\angle E = 90^\circ$, and triangle GHI has $GH = 9$, $HI = 12$, $\angle H = 90^\circ$. Are the triangles congruent? By which criterion?

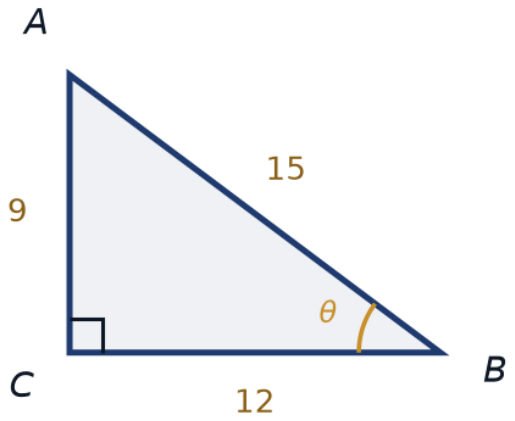
9 [Intermediate] • Target: Triangle congruence criteria

Two triangles have all three pairs of corresponding sides equal in length. Which congruence criterion applies, and are the triangles necessarily congruent?

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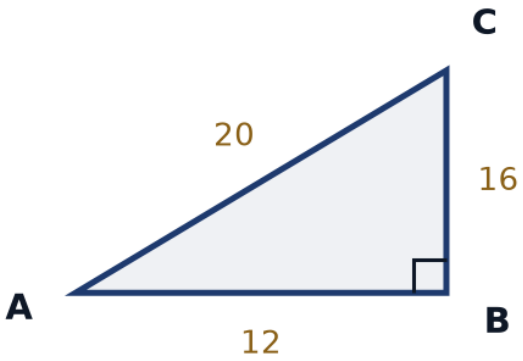
10 [Intermediate] • Target: Similar triangles and proportional reasoning

Triangle $ABC \sim$ triangle DEF, with $AB = 6$, $BC = 8$, and $DE = 9$. Find EF.



11 [Intermediate] • Target: Similar triangles and proportional reasoning

Triangle ABC shown below has $AB = 12$, $BC = 16$, and $CA = 20$. It is similar to triangle DEF, where $DE = 15$ corresponds to AB. Find EF and FD.



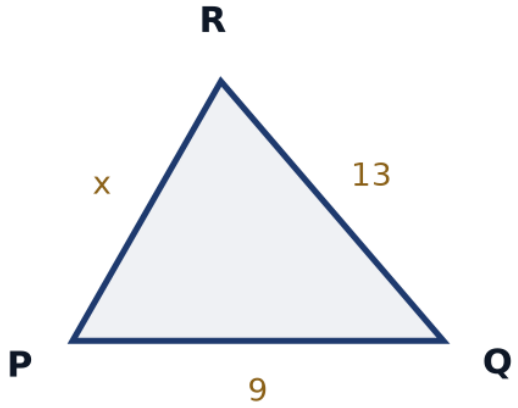
12 [Intermediate] • Target: Similar triangles and proportional reasoning

Triangle JKL has $JK = 15$ and $KL = 21$, and is similar to triangle MNO with scale factor $\frac{2}{3}$ (triangle MNO is smaller), where MN corresponds to JK. Given $MN = 10$, verify the scale factor is consistent, then find NO.

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13 [Intermediate] • Target: The Law of Cosines

In triangle PQR shown below, $PQ = 9$, $QR = 13$, and $\angle Q = 50^\circ$. Find $PR = x$, to the nearest hundredth.



14 [Intermediate] • Target: The Law of Cosines

A triangle has sides $a = 7$, $b = 8$, $c = 10$. Use the Law of Cosines to find angle C (opposite side c), to the nearest tenth of a degree.

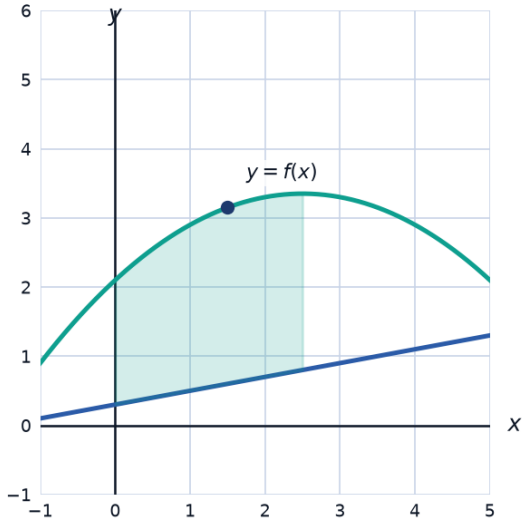
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: The Law of Cosines • Cross-Lesson: 5.01 Trigonometry

A triangle has sides $a = 6$, $b = 6$, and included angle $C = 80^\circ$. Find side c, to the nearest hundredth, and state what type of triangle this is based on the two given equal sides.

16 [Challenging] • Target: Heron's formula • Cross-Lesson: 6.01 Circles & Coordinate Geometry

Find the area of a triangle with sides 13, 14, 15 using Heron's formula.



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17 [Challenging] • Target: Heron's formula • Cross-Lesson: 7.01 Solid Geometry & Similarity

A triangular plot of land has sides 9, 10, 17. Find its area using Heron's formula.

18 [Challenging] • Target: Heron's formula • Cross-Lesson: 8.01 Statistics & Probability

A triangular sail has sides 5 m, 12 m, and 13 m. Find its area using Heron's formula, and verify using the right-triangle area formula (since 5-12-13 is a Pythagorean triple).

19 [Challenging] • Target: Interior and exterior angles of polygons • Cross-Lesson: 1.01 Advanced Algebra

Find the sum of the interior angles of a regular decagon (10 sides), and the measure of each interior angle.

20 [Challenging] • Target: Interior and exterior angles of polygons • Cross-Lesson: 2.01 Functions: Composite, Inverse & Logarithmic

Each exterior angle of a regular polygon measures 24°. How many sides does the polygon have?

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Parallel lines and transversals

An exterior angle of a triangle measures 82°, and one remote interior angle is 30°. Find the other remote interior angle.

a. 52°

b. 57°

c. 47°

d. 30°

22 [Foundational] • Target: Triangle angle sum and the exterior angle theorem

A triangle has sides $a = 6$, $b = 11$, and included angle $C = 110^\circ$. Find side c, to the nearest hundredth.

a. 13.72

b. 14.72

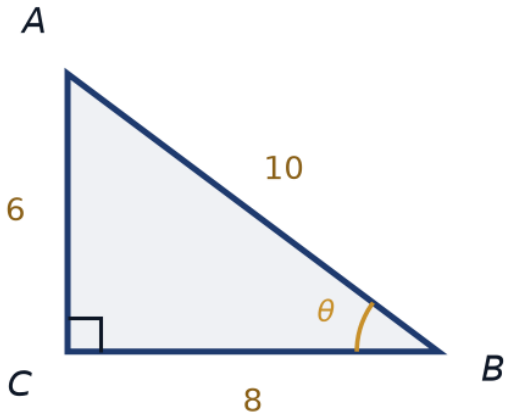
c. 14.22

d. 15.22

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23 [Foundational] • Target: Triangle congruence criteria

Find the sum of the interior angles of a regular polygon with 20 sides, and the measure of each interior angle.



a. $3060, 162$

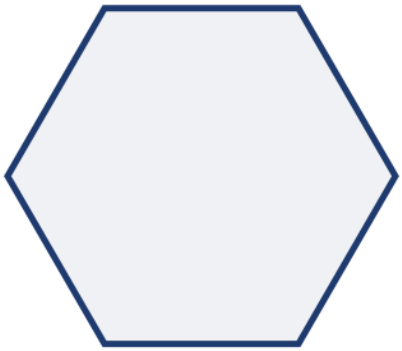
b. $3240, 162$

c. $3420, 157$

d. $3240, 167$

24 [Foundational] • Target: Similar triangles and proportional reasoning

The sum of interior angles of a hexagon is:



a. 540°

b. 720°

c. 900°

d. 360°

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25 [Foundational] • Target: The Law of Cosines

The three angles of a triangle measure $(3x - 5)^\circ$, $(3x - 10)^\circ$, and $(x + 20)^\circ$. Find the measure of the largest angle.

a. 70

b. 64

c. 62

d. 65

26 [Foundational] • Target: Heron's formula

Lines $\ell_1 \parallel \ell_2$ are cut by a transversal. Two co-interior (same-side interior) angles measure $(4x + 4)^\circ$ and 120°. Solve for x.

a. 21

b. 8

c. 20

d. 14

27 [Foundational] • Target: Interior and exterior angles of polygons

Lines $\ell_1 \parallel \ell_2$ cut by a transversal form corresponding angles of $(4x - 3)^\circ$ and $(3x + 4)^\circ$. Find the measure of the angle $(4x - 3)^\circ$.

a. 26

b. 34

c. 25

d. 20

28 [Foundational] • Target: Area of polygons

What is the sum of the interior angles of a convex polygon with 4 sides?

a. 367

b. 360

c. 368

d. 358

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: A well-designed word problem: a surveyor's triangular lots

The sum of the interior angles of a convex polygon is 2520°. How many sides does the polygon have?

a. 20

b. 16

c. 19

d. 3

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30 [Intermediate] • Target: Parallel lines and transversals

In the diagram below, line AB is parallel to line CD. Transversal line EF intersects AB at point G and CD at point H. If the measure of angle AGH is 50°, what is the measure of angle GHD?

a. 50°

b. 130°

c. 40°

d. 90°

31 [Intermediate] • Target: Triangle angle sum and the exterior angle theorem

In the diagram below, lines AB and CD are parallel. Line EF is a transversal intersecting AB at G and CD at H. If the measure of angle AGH is $(2x + 10)^\circ$ and the measure of angle GHD is $(3x - 5)^\circ$, find the value of x.

a. 15

b. 25

c. 35

d. 45

32 [Intermediate] • Target: Triangle congruence criteria

In the diagram below, triangle ABC is a right-angled triangle with the right angle at C. An altitude is drawn from C to the hypotenuse AB, meeting AB at point D. If AC = 9 and BC = 12, find the length of the altitude CD.

a. 7.2

b. 10.8

c. 8.4

d. 9.6

33 [Intermediate] • Target: Similar triangles and proportional reasoning

In the diagram below, triangle ABC is a right-angled triangle with the right angle at C. CD is the altitude to the hypotenuse AB. If AD = 4 units and DB = 9 units, find the length of the altitude CD.

a. 6 units

b. 13 units

c. 36 units

d. 7.5 units

34 [Intermediate] • Target: The Law of Cosines

Find the measure of each interior angle of a regular polygon with 18 sides.

a. 169

b. 163

c. 162

d. 160

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35 [Intermediate] • Target: Heron's formula

Each interior angle of a regular polygon measures 120°. How many sides does the polygon have?

a. 12

b. 4

c. 6

d. 14

36 [Intermediate] • Target: Interior and exterior angles of polygons

Angles S and T are supplementary. If $\angle S = 34^\circ$, find $\angle T$.

a. 139

b. 143

c. 145

d. 146

37 [Intermediate] • Target: Area of polygons

Angles M and N are complementary. If $\angle M = 69^\circ$, find $\angle N$.

a. 21

b. 25

c. 16

d. 18

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: A well-designed word problem: a surveyor's triangular lots • Cross-Lesson: 8.01 Statistics & Probability

Two lines intersect, forming vertical angles. If one angle measures 114°, what is the measure of its vertical angle?

a. 116

b. 120

c. 119

d. 114

39 [Challenging] • Target: Parallel lines and transversals • Cross-Lesson: 1.01 Advanced Algebra

Find the measure of each exterior angle of a regular polygon with 12 sides.

a. 36

b. 30

c. 32

d. 21

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40 [Challenging] • Target: Triangle angle sum and the exterior angle theorem • Cross-Lesson: 2.01 Functions: Composite, Inverse & Logarithmic

Lines $\ell_1 \parallel \ell_2$ are cut by a transversal. One angle measures $(3x + 5)^\circ$ and its corresponding angle measures 80°. Solve for x.

a. 33

b. 20

c. 25

d. 19

41 [Challenging] • Target: Triangle congruence criteria • Cross-Lesson: 3.01 Sequences, Series & Complex Numbers

A trapezoid has parallel sides 17 and 5, and height 12. Find its area.

a. 134

b. 140

c. 132

d. 136

42 [Challenging] • Target: Similar triangles and proportional reasoning • Cross-Lesson: 5.01 Trigonometry

A regular polygon has an exterior angle of 36°. How many sides does it have?

a. 6

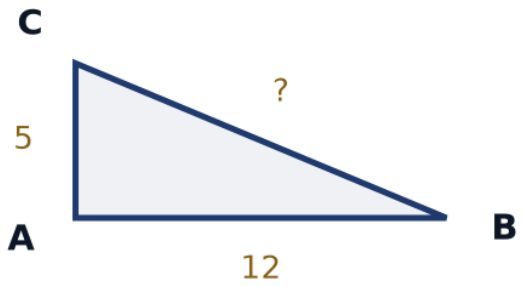
b. 10

c. 7

d. 4

43 [Challenging] • Target: The Law of Cosines • Cross-Lesson: 6.01 Circles & Coordinate Geometry

A right triangle has legs 5 and 12. Hypotenuse is:



a. 11

b. 13

c. 17

d. 15

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44 [Challenging] • Target: Heron's formula • Cross-Lesson: 7.01 Solid Geometry & Similarity

In the diagram below, line segment DE is parallel to BC. Given that AD = 6, DB = 4, and DE = 12, find the length of BC.

a. 18

b. 20

c. 30

d. 15

45 [Challenging] • Target: Interior and exterior angles of polygons • Cross-Lesson: 8.01 Statistics & Probability

In the diagram, $ABCD$ is a trapezoid with $AB \parallel DC$. The diagonals AC and BD intersect at point E. Given that $AE = 3$ cm, $EC = 6$ cm, and $BE = 4$ cm, find the length of BD.

a. 10 cm

b. 12 cm

c. 16 cm

d. 18 cm

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